IncidentIQ | System Architecture

Support roles
L1 / L2 / L3 / Admin

React + Vite UI
Role-based pages and ticket queues

FastAPI REST API
Auth, role checks, validation

SQLite database
Users, incidents, tickets, events, feedback

API search

Incident CSV
Descriptions and resolution records

Preprocessing
Normalize, clean, deduplicate, searchable text

Local searchable index
Word/char TF-IDF, local SVD, keywords

Hybrid retrieval
Metadata filters, weighted ranking, reranking

ranked incidents

Incoming query
Incident symptoms and impact

Guardrails + triage
PII filtering, classify, prioritize, route

Resolution assistance
RCA and retrieved historical evidence

Safe answer
Grounding validation and source incidents

resolution / escalation

L1 support
Create, update, assign priority, resolve

Assigned L2 handoff
L1 escalates with notes and diagnostics

Assigned L3 handoff
L2 escalates; L3 continues resolution

L3 guidance / closure
LLM guidance or local offline fallback

Evaluation
Representative query set; P@1, Hit@3/5, MRR, MAP, latency

Feedback and operations
User feedback, admin analytics, audit and notifications

Local embeddings, hybrid search, reranking and deterministic fallback run without a configured API key.